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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
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A truism in mammalian development is that fertilization is not difficult, but implantation is. About 70% of all human embryos die before birth, and the time of greatest embryonic loss is during implantation. (Cross et al., 1994). In this period, the embryonic, and extraembryonic yolk sac and placental cell lineages are determined and become essential for survival (Rappolee, 1999). In humans, failure to form a placenta properly leads to the clinically important conditions preeclampsia and intrauterine growth retardation (IUGR). Experiments have shown that perturbation of maternal or embryonic development only during the late preimplantation stages can lead to detrimental ramifications (Rutedge et al., 1994) during later development and at parturition (Kwong et al., 2000).
Hypothesis 1: Early cleavage division events are sensitive to gravity. Hypothesis 1a, the gravity-sensitive period is only during the 2-8-cell-stage when the block in embryonic development was observed (Kojima et al., 2000), and post-8-cell

Task Description:	development is normal (if embryos escape the block). Hypothesis 1b, the gravity-sensitive period also occurs during post-8-cell development after compaction. Specific aim 1. (1) Which stage of preimplantation embryonic development is gravity-sensitive? To test this, we will make daily measurements of several criteria of embryo development: (1) anabolic deviation (RNA, protein content and synthetic rates), (2) net growth rate (cell number), (3) components contributing to net growth rate (fraction in mitosis S phase, fraction in apoptosis), (4) morphogenetic events (compaction rate, cavitation rate), (5) and allocation to embryonic and placental lineages (fraction and total number inside and outside cells). Analysis of preimplantation development will be done to distinguish whether pre-compaction (2-8-cell-stage) development that is the primary period of perturbation (Kojima et al., 2000). What events are affected by microgravity? Zygotic genome activation (ZGA), compaction and cavitation will be tested for delay of initiation and quantitatively for relative strength. This will distinguish whether events during the 2-8-cell-stage block (Kojima et al., 2000) have an earlier onset. Post-compaction events such as cavitation and trophoblast lineage allocation will be tested for indication that development of the embryo is perturbed after the 2-8-cell-stage block-embryo. Hypothesis 2: Preimplantation embryos are sensitive to gravity and embryos cultured under conditions of microgravity during the (1) entire preimplantation development, (2) from 1-8-cell- stage, or (3) from post-compaction 8-cell-stage through blastocyst-stage show gravity-sensitive phenotypes. These phenotypes are observed in two post-implantation models; (1) when embryos from these three groups produce smaller litter sizes/weights after reimplantation and (2) when they produce deviant early post-implantation events measured in blastocyst or ICM outgrowth culture. Specific aim 2 (2) Are embryos that progress to the most advanced preimplantation stages (morula and blastocyst) in microgravity culture normal when reimplanted or cultured? (2a) Do embryos cultured in microgravity from (1) 1-cell-stage through blastocyst-stage, (2) from 8-cell-stage through blastocyst-stage, or (3) from 1-cell-stage through 8-cell-stage reimplant and produce normal quantity of newborns? Analyze number and weight of newborn mice produced from embryos culture in microgravity for parts or all of preimplantation development. If number or weight is decreased, do follow-up studies on size of embryos during mid- and late-uterine growth periods. (2b) Do blastocyst cultured in microgravity (1) from 1-cell to blastocyst-stage, or from 8-cell-stage through blastocyst-stage yield the normal placental/trophoblast and yolk sac/endoderm growth and differentiation events in blastocyst outgrowth culture at normal gravity? Analysis of blastocyst outgrowth culture using markers for trophoblast and endoderm proliferation, differentiation and apoptosis will be done to test for effects of microgravity on preimplantation embryos.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will inform us whether gravity has a role during mammalian embryogenesis as the embryo prepares to set up the head-to-tail, back-to-belly, and left-to-right axes. In addition, we will test whether embryonic movement set up during microgravity simulation is itself stressful and deleterious to embryogenesis before implantation into the uterus. In the first year of the grant, we have found that there is a profound and complete block of embryos during the step at which the first placental epithelium (and the first epithelium of any king)is made.
Task Progress:	HARV was run (7.5RPM) with a stationary media microdroplet at 1.2dynes/cm2, about 1/10th the amount of shear stress sensed by arteriolar endothelial cells. DV digital movies of the stationary droplet and the embryos inside the droplet were recorded and sent to our collaborator, Dr Shen (a NASA-funded laminar flow scientist), who solved a free body analysis of the motion and amount of shear stress on the embryos. This amount of shear stress is about 10-100x the amount of shear stress sensed by the embryo during transit from the ovary through the oviduct, into the uterus. The acellular protein coat, the Zona pellucida protects the embryo from the 1-cell zygote immediately after fertilization (in the upper oviduct) until the blastocyst stage 4.5 days later (E4.5), when the embryo hatches from the Zona pellucida and implants into the uterus. Since the embryos attach and become stationary after attachment, while fluids in the uterine lumen continue to flow over the embryo surface, we measured SAPK/JNK phospho at E4.5 by Western blot and found it high. The removal of the Zona pellucida leads to a more rapid induction of embryo lethality, suggesting a function of the Zona pellucida is to protect the embryo during transit through the oviduct and uterus. We also found that mouth pipetting embryos increases both SAPK/JNK phospho and c-Fos in a dose-dependent day. We are currently developing means to make the embryos metabolically inactive, so that embryos do not change SAPK/JNK phospho amounts due to artifactual treatment during isolation. Another variable arising during shear stress or microgravity simulation due to movement, is the loss of hypoxia, a known mitogen to the placental stem cells in the outer epithelium of the preimplantation embryo. However, when the RCCS1/HARV protocols were done in a hypoxic incubator, the shear stress and microgravity-simulated groups were not affected (although there was a significant delay in the rotational controls). That shear stress might lead to activation of the essential “implantation phenotype” has recently been tested. The implanting embryos must differentiate (placental) trophoblast stem (TS) cells to make trophoblast giant (TG) cells that signal directly to the mother during early gestation that the embryo is present. This signaling maintains the corpus luteum of the ovary and maintains progesterone section by the corpus luteum essential for embryonic survival. The signal is hCG is human (ligand of the early pregnancy test), and placental lactogen (PL)-1 in the mouse. During differentiation, Inhibitor of Differentiation (Id)-2 (basic Helix loop helix dominant negative) must be suppressed and Heart and mesoderm differentiation inducer (HAND1, basic helix loop helix transcription factor that regulates the promoter of PL-1) must be stabilized. We have found that stress on the E3.5 embryo leads to fast Id2 export from the nucleus of TS cells, leaving only HAND1 in the nucleus. In TS cells, (hyperosmolar) stress causes Id2 export and then loss of total protein in a SAPK/JNK-independent manner. Stress also causes stabilization and increase in HAND1 in a SAPK/JNK-dependent manner. PL-1 is stress-induced in manner partially dependent on SAPK/JNK. We are currently pursuing the other stress enzymes that contribute to stress sensing in the outer placental cells of the implanting embryo and we are also testing for whether the shear stress regulates the induction of PL-1 in the implanting embryo after hatching and loss of the Zona Pellucida. Future studies It will be important to create a dose-response for shear stress for embryos using the RCCS1. This could be done by using different densities and viscosities of oil that would allow a range of angular velocities of the HARV and a range of resulting shear stress magnitudes. Since embryos are not normally adhesive, typical shear stress protocols (like those for endothelial cells) can’t be used. The RCCS1 is unique in its ability to generate shear stress on embryos in the non-adherent state. A dose-response would allow a better understanding of the kind of shear stress signaling mechanisms in the embryo and better model the in vivo milieu. It will be important to perform an experiment to test for the effects of microgravity in spaceflight. At the start of this project, we anticipated homeotic

	effects of microgravity as frog and chich require normal gravity with a constant force to pattern the embryo. Homeotic effects are still a possibility in mammalian embryos, but it is clear that microgravity simulation and shear stress create a high level of SAPK and cell suicide that is not compatible with embryonic life. It is important to verify these findings in spaceflight. This work was supported by the following grants NASA (NAG2-1503), NIH (R01 HD40972A). Additional help was kindly given by Dr Josh Zimmerberg of NICHD, who provided 2 additional RCCS1 and 10mL HARVS. These additional RCCS1s allowed us to control experiments while developing more data points/experiment for shear stress time courses and SAPK/JNK inhibitor studies.
Bibliography Type:	Description: (Last Updated: 09/23/2009)
Abstracts for Journals and Proceedings	Xie, Y, Wang, F, Puscheck, EE, and DA Rappolee "Shear stress is lethal to preimplantation mouse embryos by induction of SAPK/JNK-mediated apoptosis (TUNEL) A second function of the Zona pellucida is to prevent shear stress associated lethality." Gravit. Space Biol. Bull. 18:60 , Nov-2004
Abstracts for Journals and Proceedings	Rappolee, DA, Xie, Y, Wang, Y, Wang, F, Wygle D, Shen, H., and EE.Puscheck, "A major effect of simulated microgravity on preimplantation mouse development is lethality." Gravit. Space Biol. Bull. 18:28 , Nov-2004
Articles in Peer-reviewed Journals	Yingchun Wang, Elizabeth E. Puscheck, Dana L Wygle, Jennifer J Lewis, Anna B Trostinskaia, Fangfei Wang, and Daniel A Rappolee "Increases in phosphorylation of SAPK/JNK and p38MAPK correlate negatively with mouse embryo development after culture in different media" Fertility and Sterility 2005 Apr;83:1144-1154. , Apr-2005
Articles in Peer-reviewed Journals	Zhong W, Sun T, Wang QT, Wang Y, Xie Y, Johnson A, Leach R, Puscheck EE, Rappolee DA "SAPKgamma/JNK1 and SAPKalpha/JNK2 mRNA transcripts are expressed in early gestation human placenta and mouse eggs, preimplantation embryos, and trophoblast stem cells. " Fertility and Sterility. 2004 Oct;82:1140-1148 , Oct-2004
Articles in Peer-reviewed Journals	Liu, J., Puscheck, EE, Wang, F, Trostinskaia, A, Barisic, D, Maniere, G, Wygle, D, Zhong, W, Rings EHHM, and DA Rappolee "Serine-threonine kinases and transcription factors active in signal transduction are detected at high levels of phosphorylation during mitosis in preimplantation embryos and trophoblast stem cells" Reproduction 2004 Nov;128: 643-654. , Nov-2004